

Problem-Solution Fit

Date	1 July 2026
Team ID	SWTID-2026-4523
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

<u>1.CUSTOMER SEGMENT(S)</u> <u>[CS]</u> Banks, financial institutions, and credit card companies are the primary users. The system also benefits loan officers and applicants by providing faster approval decisions.	<u>6.CUSTOMER</u> <u>LIMITATIONS EG:</u> <u>BUDGET, DEVICES</u> <u>[CL]</u> Some organizations have budget constraints and limited technical expertise. Incomplete applicant information and concerns about data privacy can also affect the approval process.	<u>5.AVAILABLE</u> <u>SOLUTIONS PROS &</u> <u>CONS [AS]</u> Manual verification is reliable but slow. Rule-based systems are faster but less flexible. Machine learning provides quicker and more accurate decisions.
<u>2.PROBLEMS/PLANS</u> <u>+ITS FREQUENCY [PR]</u> Banks receive a large number of credit card applications every day. Manual verification is time-consuming, prone to errors, and often results in delayed approval decisions.	<u>9. PROBLEM</u> <u>ROOT/CAUSE [RC]</u> The primary cause of delayed approvals is the dependence on manual verification and traditional decision-making processes. Increasing application volumes further reduce processing efficiency.	<u>7. BEHAVIOUR</u> <u>+ ITS INTENSITY[BE]</u> Customers expect instant and transparent approval decisions. Financial institutions require a reliable system that can efficiently process a high volume of applications every day.

3. TRIGGERS TO ACT [TR]

The growing number of applications and the need for quick, accurate decisions encourage organizations to adopt an automated credit approval system.

4. EMOTIONS

Before using the system, applicants may feel anxious due to long waiting times, while bank staff experience heavy workloads. After implementation, applicants receive quicker responses, and employees can process applications more efficiently.

10. YOUR SOLUTION [SL]

A Machine Learning-based Credit Card Approval Prediction System developed using Python, Flask, and Scikit-learn. It predicts approval decisions quickly, accurately, and consistently while reducing manual effort.

8. CHANNELS OF BEHAVIOUR [CH] **ONLINE**

Online: Bank websites, mobile banking applications, and customer portals are commonly used for submitting credit card applications.

Offline: Applications can also be submitted through bank branches and customer service centers.

OFFLINE